An Unofficial Compendium of Worked Solutions to the Problems in *The Method of Coordinates*

by I.M. Gelfand, E. G. Glagoleva & A. A. Kirillov

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Abstract

This book is an independent and unofficial collection of worked solutions to problems inspired by the book *The Method of Coordinates* by I. M. Gelfand, E. G. Glagoleva and A. A. Kirillov, first published in 1967. The original text presents a wide range of problems in elementary algebra, some accompanied by solutions and others left as exercises for the reader.

The aim of this book is to provide clear, logically consistent solutions to all problems addressed, written in a style accessible to motivated students and self learners. Proofs have been verified primarily by hand, and in some cases supported by small computational checks written in Scheme (a dialect of Lisp) or in Python. Large language models were used only as auxiliary tools, primarily for generating figures in LaTeX and for reviewing exposition.

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Unofficial companion.

This document is an **independent, unofficial collection of worked solutions** to problems from *The Method of Coordinates* by I. M. Gelfand , E. G. Glagoleva and A. A. Kirillov (first published 1967).

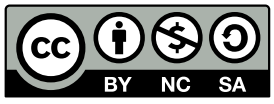
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Project repository:
https://github.com/deepak-venkatesh/gelfand-method-coordinates

For

My Daughter and Son

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Working Notes

Challenging Problems for Middle Schoolers:

Problem Numbers:

Typographical errors in:

Problem Numbers:

Part I

Part 1

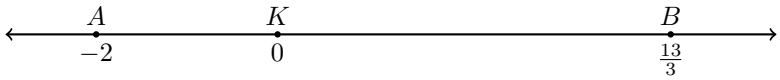
1 Chapter 1: The Coordinates of Points on a Line

1.1 The Number Axis

Exercise 1

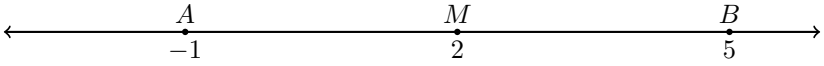
(a)

The points are marked below. Note that $\frac{13}{3}$ lies somewhere between 4 and 5.



(b)

The two points are 3 units to the right and 3 units to the left of M as shown in the line below.



Exercise 2

(a)

Going by the convention that the right side of the number line indicates larger nummber, in this exercise $a > b$.

(b)

On the right are the larger numbers

- $A(-3)$
- $B(4)$
- $B(4)$
- $A(3)$

Exercise 3

It is easy to say that $A(a)$ is on the right of $B(-a)$ but this assumes that a is a positive number. So this could be true but the other case is that if a were a negative number then $A(a)$ would be a negative number and $B(-a)$ would be a positive number, thus $B(-a)$ would be on the right.

The third case is that $a = 0$ then both A and B would be 0 and they would be at the same point on the number line.

Exercise 4

(a)

If $x = 0$, both $M(x)$ and $N(2x)$ will be at the same point on the number line.

If $x > 0$, $N(2x)$ will be on the right of $M(x)$ on the number line.

If $x < 0$, $M(x)$ will be on the right of $N(2x)$ on the number line.

(b)

In this problem no matter whether c is 0, positive or negative $B(c + 2)$ will always be on the right side of $A(c)$.

(c)

In this problem we have 3 scenarios.

If $a = 0$ then $A(x)$ and $B(x - a)$ are the at the same point in the number line.

If $a > 0$ then $A(x)$ will be on the right of $B(x - a)$.

If $a < 0$ then $B(x - a)$ will be on the right side of $A(x)$ since the negative number multiplied by $-$ will give a positive number.

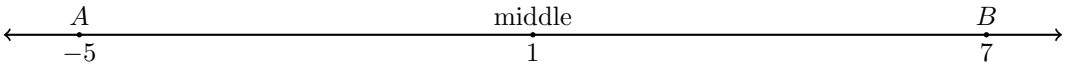
(d)

If $x = 0$ then both $A(x)$ and $B(x^2)$ would be at the same point in the number line.

But in any other case whether x is positive or negative x^2 will always be greater (and positive) that x. Thus $B(x^2)$ will always be on the right in these two cases.

Exercise 5

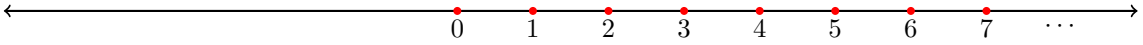
The space or distance between the two points is 12 (-5 to 0 and 0 to 7). The middle point will be 6 units from -5 on the right which is at point 1.



Exercise 6

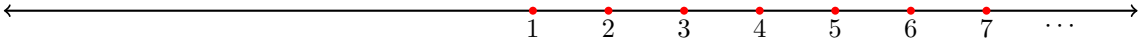
(a)

Whole numbers will include number 0.



(b)

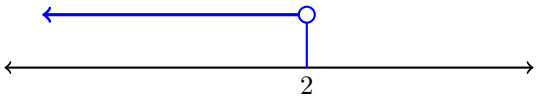
Positive numbers will not include number 0.



Exercise 7

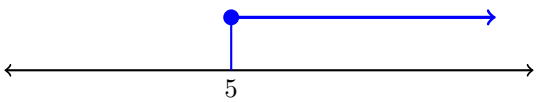
(a)

For $x < 2$



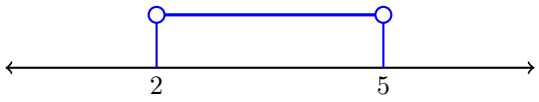
(b)

For $x \geq 5$



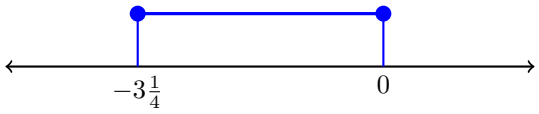
(c)

For $2 < x < 5$



(d)

For $-3\frac{1}{4} \leq x \leq 0$



1.2 The Absolute Value of a Number

Exercise 1

There will be 3 cases:

Case 1: $x = 0$. In this we are dividing by 0 so the division is undefined.

Case 2: $x < 0$. The expression is $\frac{x}{-x} = -1$. It is -1.

Case 3: $x > 0$. Here it is simply 1

Exercise 2

(a) The square will always be positive. Hence the term a^2 with or without modulus will always be positive. We can write it as.

$$|a^2| = a^2$$

(b) The term will always be positive since $a > b$.

$$|a - b| = a - b$$

(c) In this case the term will be negative since $a < b$.

$$|a - b| = -(a - b) = b - a$$

(d) a is negative so negative of it will be positive, therefore we will get the below.

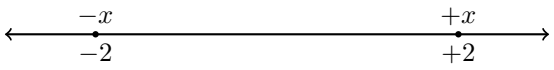
$$|-a| = a$$

Exercise 3

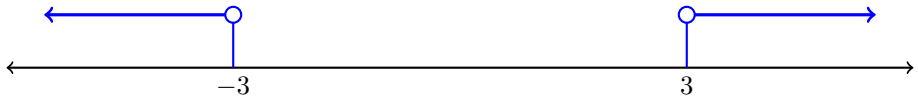
We know that the term $x - 3$ will be non negative. So we can say that $x \geq 3$.

Exercise 4

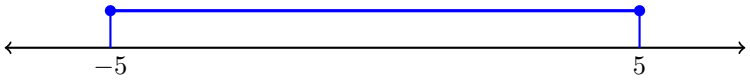
(a) $|x| = 2$ will be +2 and -2



(b) The solution for $|x| > 3$ will be $(-\infty, -3) \cup (3, \infty)$

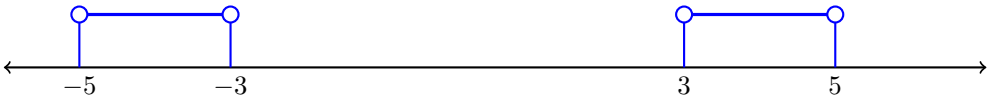


(c) The solution for $|x| \leq 5$ will be $[-5, 5]$



(d) This means that x is between 3 and 5 when x is positive and it will be between -5 and -3 when negative.

The solution set is $(-5, -3) \cup (3, 5)$



(e) There will be two cases for $|x - 2|$ as shown below.

$$|x - 2| = \begin{cases} x - 2, & x \geq 2 \\ 2 - x, & x < 2 \end{cases}$$

For the first case $x \geq 2$

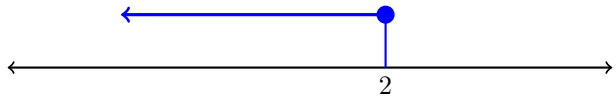
$$\begin{aligned} |x - 2| &= 2 - x \\ x - 2 &= 2 - x \\ 2x &= 4 \\ x &= 2 \end{aligned}$$

For the second case $x < 2$

$$\begin{aligned} |x-2| &= 2-x \\ -x+2 &= 2-x \\ 2-x &= 2-x \end{aligned}$$

This will be true for all $x < 2$

Thus the solution will be $x \leq 2$ and the set will be $(-\infty, 2]$



Exercise 5

(a) A point which is 2 units from 3. This gives us 5 and -1.

From the equations we can also solve it as:

$$\begin{aligned} |x-2| &= 3 \\ x-2 &= 3 \\ x &= 5 \end{aligned}$$

and,

$$\begin{aligned} |x-2| &= 3 \\ -x+2 &= 3 \\ x &= -1 \end{aligned}$$

(b) Let us reason out verbally. The addition of two numbers (the absolute values make them positive) is 1. Now since both are positive these values have to be between 0 and less than 1 i.e. $[0, 1]$. Next let us look at the two absolute values and see what value x can take for each of them.

Case: $|x+1| = 0$

$$\begin{aligned} |x+1| &= 0 \\ x &= -1 \end{aligned}$$

$x = -1$ satisfies the main equation $|x+1| + |x+2| = 1$

Case : —x + 1— = 1

$$|x+1| = 1$$
$$x = 0, -2$$

But we notice that only $x = -2$ satisfies the main equation but not $x = 0$

Case 3: $|x+2| = 0$

$$|x+2| = 0$$
$$x = -2$$

This also satisfies the main equation.

Case 4: $|x+2| = 1$

$$|x+2| = 1$$
$$x = -1, -3$$

But we notice that only $x = -1$ satisfies the main equation but not $x = -3$

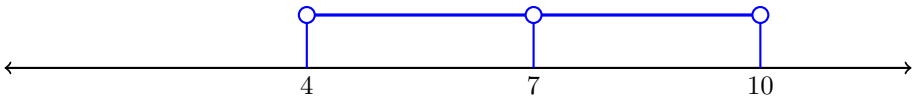
Therefore our critical points are $x = -2$ and $x = -1$. Next we observe and number to the left of -2 will not satisfy the main equation and any number on the right of -1 will not satisfy either. All numbers between and inclusive of -1 and -2 satisfies the main equation. Therefore, the solution is $-2 \leq x \leq -1$.

1.3 The Distance Between Two Points

Exercise 1

(a)

This includes all numbers from 4 to 7 and from 7 to 10. Both 4 and 10 are not included. We will write this as $(4, 7) \cup (7, 10)$



(b)

$|x-2| > 1$ can be solved with two cases

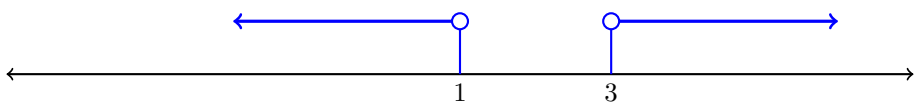
Case 1:

$$|x-2|>1$$
$$x-2>1$$
$$x>3$$

Case 2:

$$|x-2|>1$$
$$-x+2>1$$
$$x<1$$

The solution is $(-\infty,1)\cup(3,\infty)$



(c)

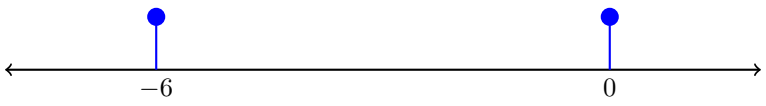
We can take two cases here too.

Case 1:

$$|x+3|=3$$
$$x+3=3$$
$$x=0$$

Case 2:

$$|x+3|=3$$
$$-x-3=3$$
$$x=-6$$



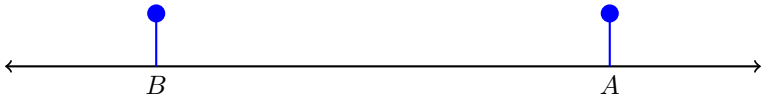
Exercise 2

There are three sub cases essentially - A and B are same, $A > B$, and $A < B$.

Case 1: $A = B$

Here $A(x_1) = B(x_2)$, thus mid point is x_1 or x_2.

Case 2: $A > B$



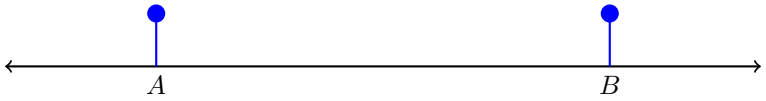
The distance between the two points is $\frac{x_1 - x_2}{2}$.

So the mid point is

$$x_2 + \frac{x_1 - x_2}{2}$$

$$\frac{x_1 + x_2}{2}$$

Case 2: $A < B$



The distance between the two points is $\frac{x_2 - x_1}{2}$.

So the mid point is

$$x_1 + \frac{x_2 - x_1}{2}$$

$$\frac{x_1 + x_2}{2}$$

Now if we look carefully that all three cases essentially give the mid point as $\frac{x_1 + x_2}{2}$.

Exercise 3

2 Chapter 2: The Coordinates of Points in the Plane

2.1 The Coordinate Plane

2.2 Relations Connecting Coordinates

2.3 The Distance Between Two Points

2.4 Defining Figures

2.5 We Begin to Solve Problems

2.6 Other System of Coordinates

3 Chapter 3: The Coordinates of a Point in Space

3.1 Coordinate Axes and Planes

3.2 Defining Figures in Space

Part II

Part 2

4 Chapter 1: Introduction

4.1 Some General Considerations

4.2 Geometry as an Aid in Calculation

4.3 The Need for Introducing Four-Dimensional Space

4.4 The Peculiarities of Four-Dimensional Space

4.5 Some Physics

5 Chapter 2: Four-Dimensional Space

5.1 Coordinate Axes and Planes

GitHub Repository:
https://github.com/deepak-venkatesh/gelfand-method-coordinates

5.2 Some Problems

6 Chapter 3: The Four-Dimensional Cube

6.1 The Definition of the Sphere and the Cube